Optimization for Machine Learning

Bogdan Savchynskyy

Admin Stuff

- Language: English (all the terminology and books are in English)

  You may ask questions in German.
- Lecturer: Bogdan Savchynskyy
- Exercises: Alexander Kirillov
- Staff Email: bogdan.savchynskyy@tu-dresden.de
- Course page: http://cvlab-dresden.de/courses/opt-machine-learning/
- Slides+script - online
- Announcements: online
- Books:

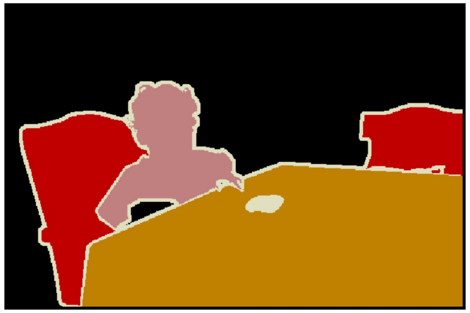
  Sebastian Nowozin and Christoph H. Lampert.
  Structured Learning and Prediction in Computer Vision

  References to books, conference and journal articles after each block

ML: Main Concepts

Observation, o

Inference



Object state, x

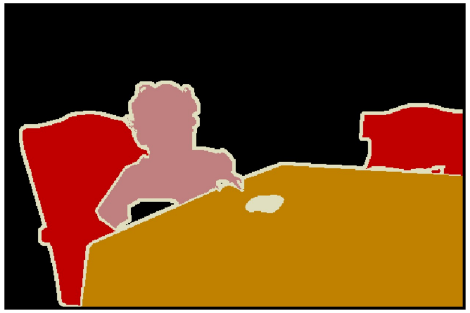
Model $p(o, x)$ Probability theory

Decision strategy $$x' = \arg\max_x p(x/o)$$ Theory of statistical decisions

Optimization

ML: Main Concepts

Observation, o



Object state, x

Learning $\longrightarrow$ Parameters, w

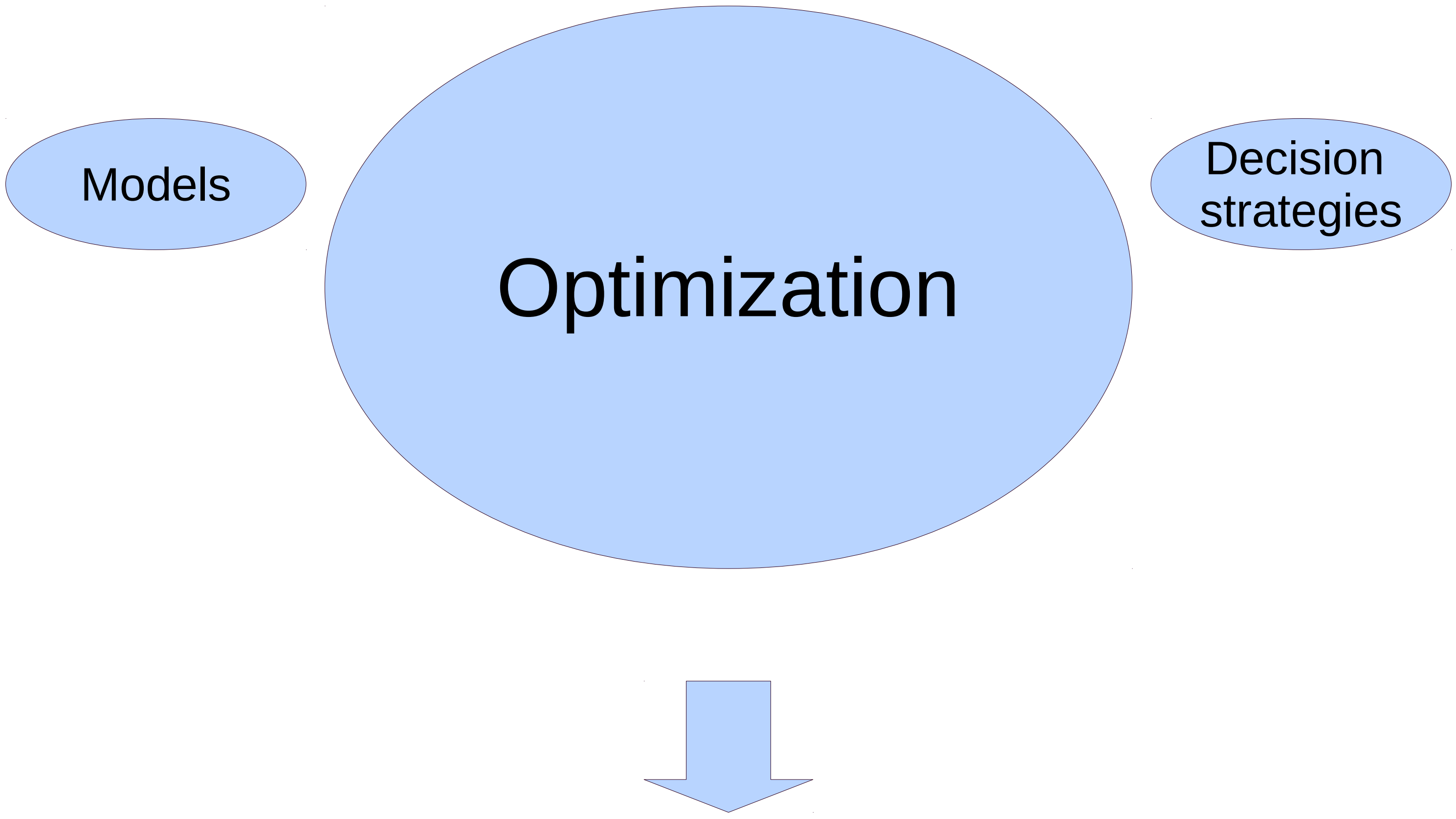
$$p_w(o,x)$$

Model $\quad p_w(o,x) \quad$ Probability theory

Learning strategy $\quad \min_w \left(x - \arg\max_{x'} p_w(x'/o)\right) \quad$ Theory of statistical decisions

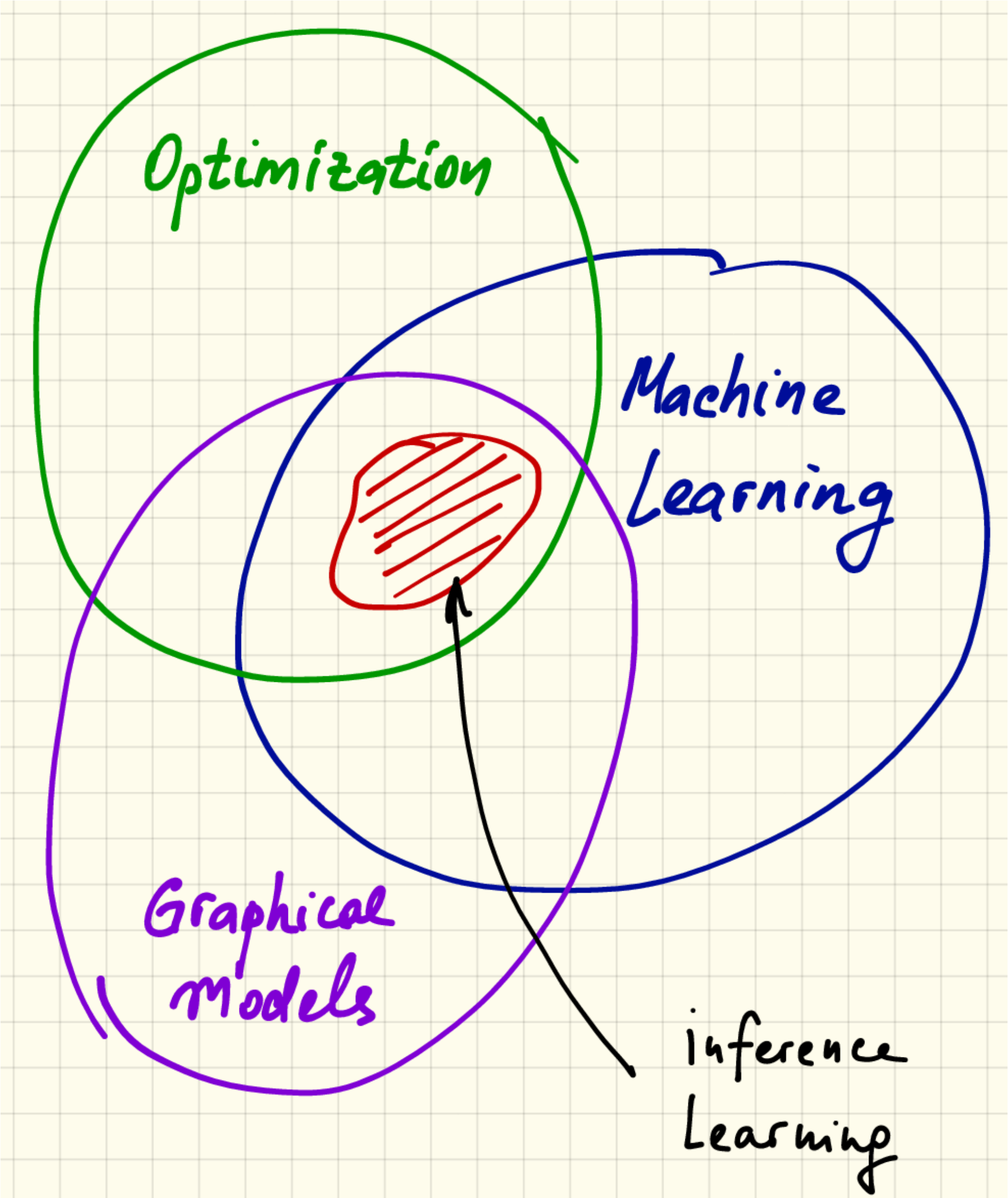
Optimization

Accents of this course



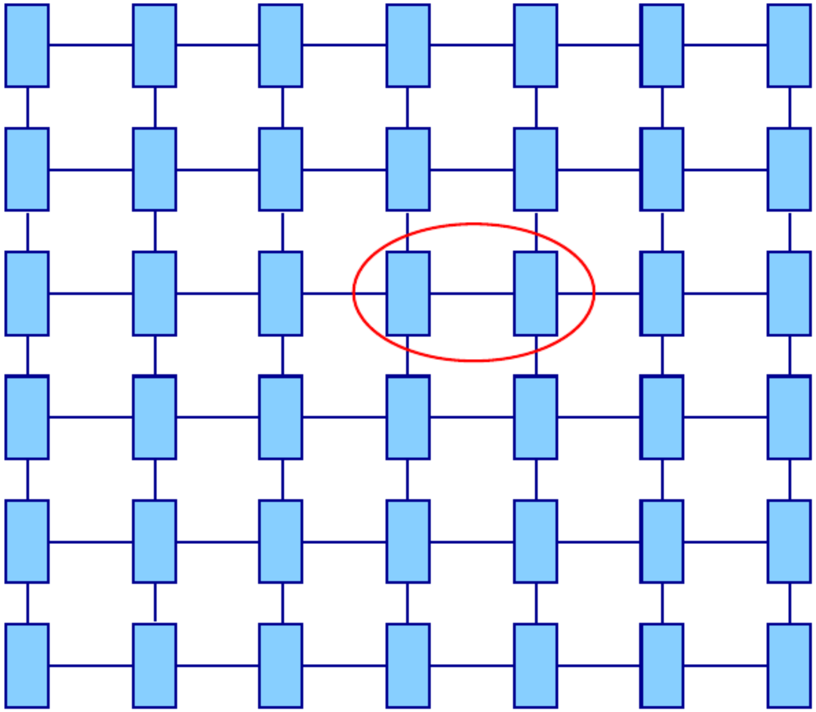
Trade-off between model expressiveness and inference/learning efficiency

Accents of this course

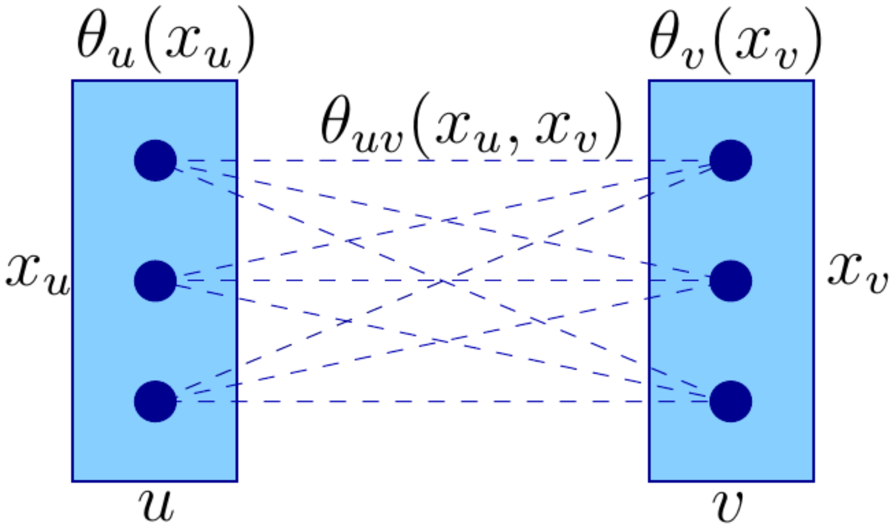


Graphical Models, Inference

$$x^* = \min_{\overline{x} \in X_V} E_\theta(\overline{x}) := \min_{\overline{x} \in X_V} \sum_{v \in V} \theta_v(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}(x_u, x_v)$$

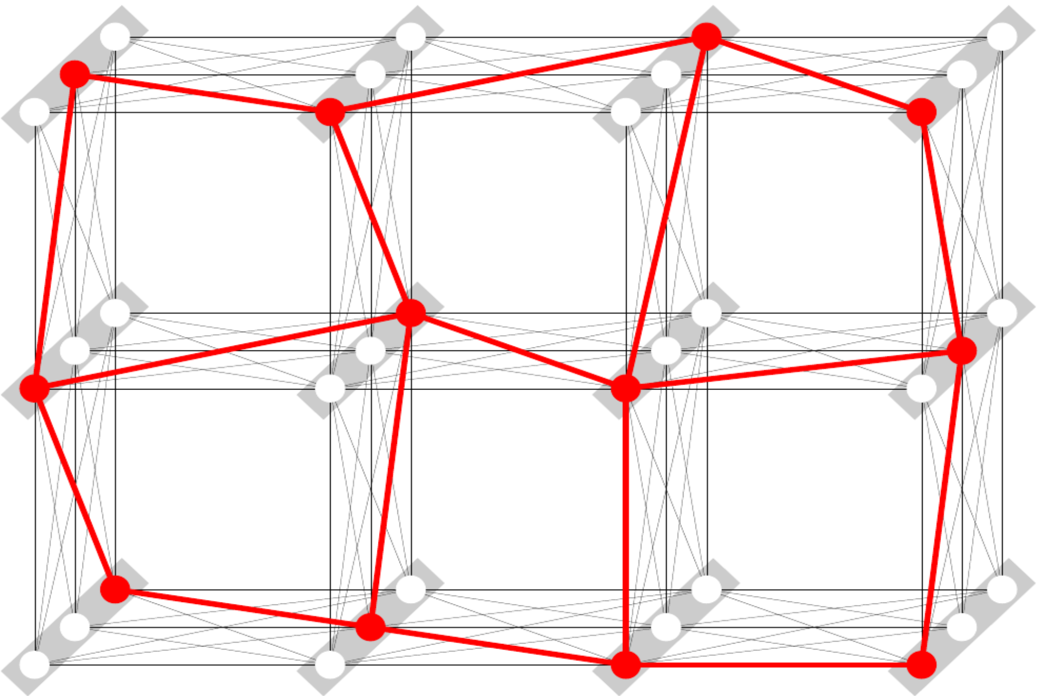


graph $(\mathcal{V}, \mathcal{E})$



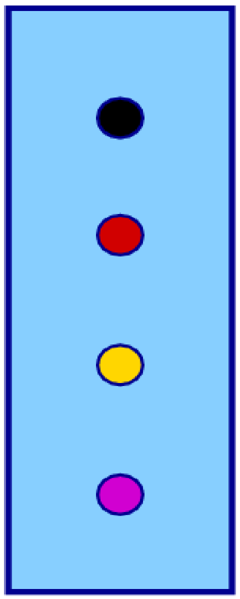
Labeling

$$x^* = \min_{\overline{x} \in X_V} E_\theta(\overline{x}) := \min_{\overline{x} \in X_V} \sum_{v \in V} \theta_v(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}(x_u, x_v)$$



Picture: T Werner. A Linear Programming Approach to Max-sum Problem: A Review

Example: Image Segmentation

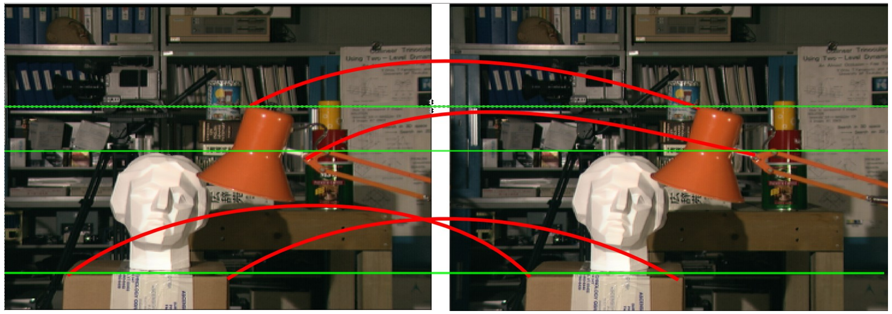


Labels = Segment names, V – set of pixels

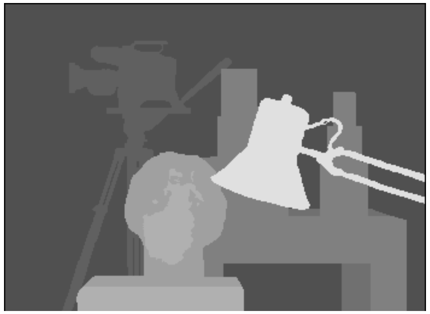
$$\theta_v(x_v)$$ - data term

$$\theta_{uv}(x_u, x_v) = \lambda |x_u \neq x_v|$$

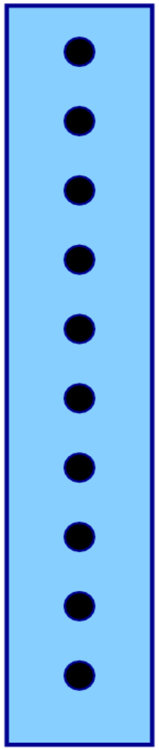
Example: Calibrated stereo reconstruction



Left image   Right image



Disparities



Labels = 1D disparities, V – set of pixels

$$\theta_v(x_v)$$ - data term

$$\theta_{uv}(x_u, x_v) = \lambda |x_u - x_v|$$ - regularizer

Pictures: Middlebury Benchmark

Example: Calibrated stereo reconstruction

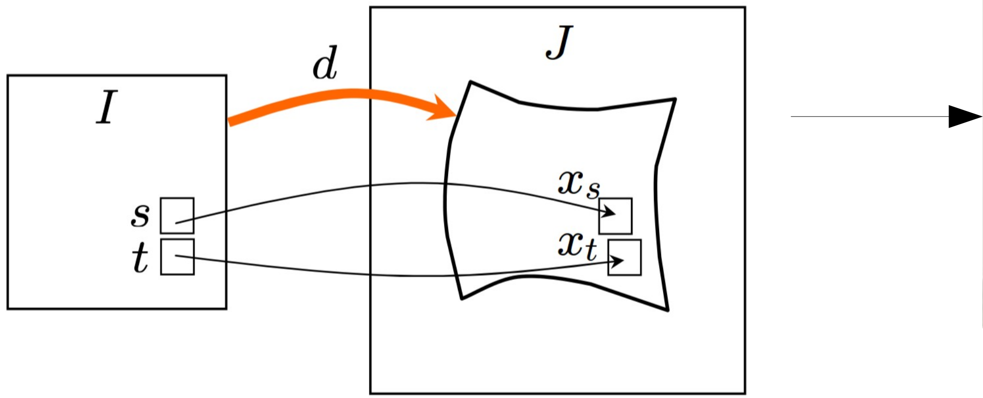
Stereo pair

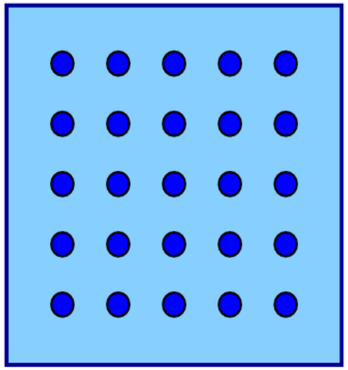
Local reconstruction

Globally consistent reconstruction

Picture: courtesy of D. Schlesinger

Example: Optical flow/Non-rigid matching





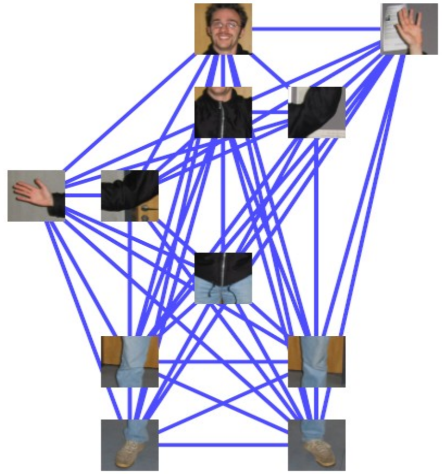
Labels = 2D disparities, V – set of pixels

$$\theta_v(x_v)$$ - data term

$$\theta_{uv}(x_u, x_v) = \lambda \|x_u - x_v\|^2$$ - regularizer

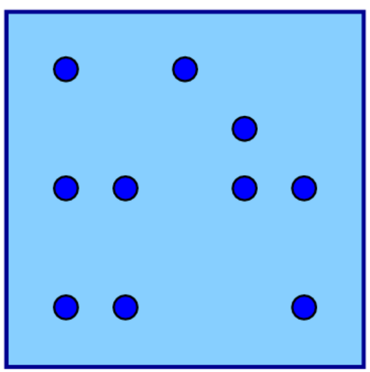
Pictures: http://www.irtc.org.ua/image/

Example: Part-Based Object Detection



Labels = 2D coordinates (sparse),

V – set of parts, Tree-structure

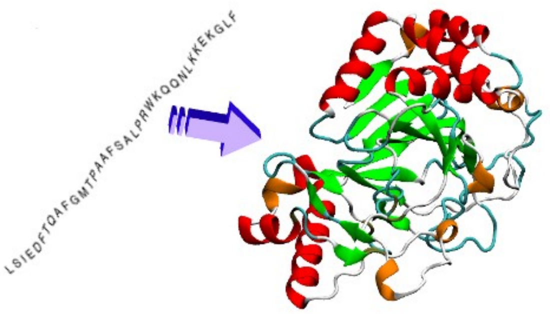


$$\theta_v(x_v)$$ - data term

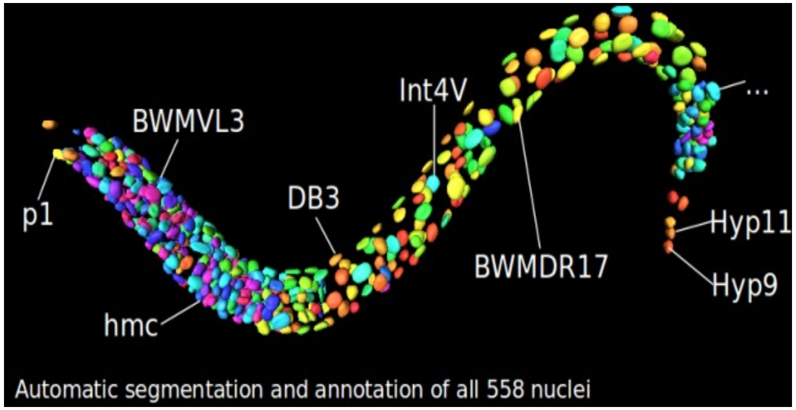
$$\theta_{uv}(x_u, x_v)$$ - geometric prior

Pictures: Bergtholdt, M. and Kappes, J. H. and Schmidt, S. and Schnörr, C.: A Study of Parts-Based Object Class Detection Using Complete Graphs

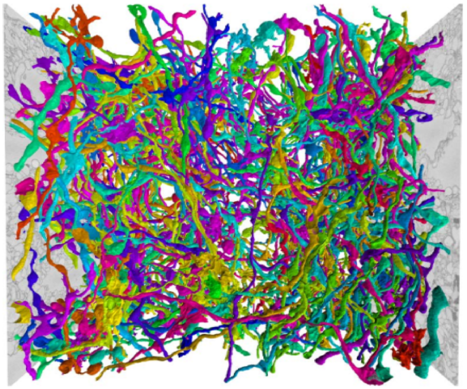
Other examples...



Protein folding



Reconstruction of structure of simple organisms



Segmentation of neurons

Panorama stitching

A number of other examples at CV-II lectures!

Statistical Meaning

$$x^* = \min_{\overline{x} \in X_V} E_\theta(\overline{x}) := \min_{\overline{x} \in X_V} \sum_{v \in V} \theta_v(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}(x_u, x_v)$$

Gibbs distribution (factorized):

$$p_\theta(\overline{x}) = \frac{1}{Z(\theta)} \exp(-E(\overline{x}))$$

$$= \frac{1}{Z(\theta)} \prod_{v \in V} \Theta_v(x_v) \prod_{uv \in \mathcal{E}} \Theta_{uv}(x_u, x_v)$$

$$\Theta_v(x_v) = \exp(-\theta_v(x_v)) \quad \Theta_{uv}(x_u, x_v) = \exp(-\theta_{uv}(x_u, x_v))$$

$$Z(\theta, T) = \sum_{\overline{x} \in X_V} \exp(-E(\overline{x})) - \text{partition function}$$

Statistical Meaning

$$x^* = \min_{\overline{x} \in X_V} E_\theta(\overline{x}) := \min_{\overline{x} \in X_V} \sum_{v \in V} \theta_v(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}(x_u, x_v)$$

$$p_\theta(\overline{x}) = \frac{1}{Z(\theta)} \exp(-E(\overline{x}))$$

Maximum a'posteriori decision (MAP-inference):

$$\arg\max_{\overline{x} \in X_V} p_\theta(\overline{x}) = \arg\min_{\overline{x} \in X_V} E_\theta(\overline{x})$$

Relation to other formalisms

Related areas:

- Hidden Markov Models
- Finite Automata and Formal languages
- Constraint Satisfaction Problems
- Markov Random Fields

Other names in other communities:

Maximum likelihood estimation (MLE) inference
Weighted constraint satisfaction problem
Constraint optimization problem
Energy minimization for graphical models

References

Segmentation: Pascal VOC 2010

Stereo, Panorama: Middlebury Benchmark

Non-rigid matching:

- Shekhovtsov, I. Kovtun, V. Hlavac:
  Efficient MRF Deformation Model for Non-Rigid Image Matching
- Facial expression synthesis project

Part-based object detection:

Bergtholdt, M. and Kappes, J. H. and Schmidt, S. and Schnörr, C.:
A Study of Parts-Based Object Class Detection Using Complete Graphs

Other datasets: OpenGM2 Benchmark, Worm, Neurons, Protein folding